**TECHCRUSH**

Cybersecurity Track – Cohort 8 Capstone Project

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# PROJECT 3

## SIEM Deployment
## Security Operations Report

Splunk SIEM • Log Ingestion • Attack Simulation • Detection Engineering
Linux and Windows Endpoints • MITRE ATT&CK • Formal Incident Report

## GROUP 18

Cohort 8

*Controlled academic cyber-range exercise on isolated VirtualBox virtual machines*
*Group members are listed on the next page.*

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## Executive Summary

This report documents the end-to-end deployment and validation of a Security Information and Event Management (SIEM) environment for a newly formed Security Operations Center (SOC). Splunk Enterprise 10.4.3 was deployed as the central manager on Kali Linux, while Ubuntu Server and Windows 11 Pro were provisioned as monitored endpoints. Splunk Universal Forwarders were configured to transmit Linux authentication/syslog telemetry and Windows Application, Security, and System event logs to the central Splunk instance over TCP port 9997.

After baseline log ingestion was confirmed, three controlled attack scenarios were executed against the lab endpoints: (1) creation of an unauthorized local administrator account, (2) repeated SSH password-guessing attempts against the Linux endpoint, and (3) clearing of the Windows Security audit log. Splunk successfully ingested the resulting telemetry and supported detection engineering for all three scenarios. A tuned SSH brute-force rule identified five failed attempts in a five-minute window, a Windows account-creation rule detected Event ID 4720 for the test account, and a log-clearing rule detected Windows Event ID 1102.

**Outcome**
The capstone objectives were met: a central Splunk SIEM was deployed, both Linux and Windows endpoints forwarded the required telemetry, three attack vectors were simulated in a controlled lab, and detection rules plus a formal incident triage record were produced.

## Project Objectives

- Deploy a central Splunk Manager and establish a functioning SOC search/triage environment.
- Connect a minimum of one Linux endpoint and one Windows endpoint with Splunk Universal Forwarders.
- Ingest /var/log/auth.log and /var/log/syslog from Linux, and Application, Security, and System Event Logs from Windows.
- Simulate at least three controlled attack vectors and confirm that the resulting telemetry is searchable in Splunk.
- Build practical detections and tune at least one rule to reduce noise while preserving useful signal.
- Produce a formal incident report with MITRE ATT&CK mapping, raw-log proof, timeline, and remediation recommendations.

## Scope and Safety

All activities were performed only against virtual machines created and controlled for this capstone. No external or production systems were targeted. Attack simulation was limited to the isolated VirtualBox lab network.

**Report Structure**

**1. Architecture Topology**

2. Infrastructure Deployment and Log Ingestion

3. Proof of Ingestion

4. Controlled Attack Simulation

**5. Detection Engineering and Rule Tuning**

**6. Formal Incident Response Report**

7. Technical Remediation Recommendations

8. Conclusion

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## 1. Architecture Topology

The SIEM lab was deployed in Oracle VirtualBox using a shared NAT Network named NatNetwork. The network used the 10.0.2.0/24 address space. Kali Linux served as both the Splunk Manager and SOC security workstation. The Ubuntu and Windows systems acted as monitored endpoints and transmitted telemetry to Splunk over TCP port 9997.

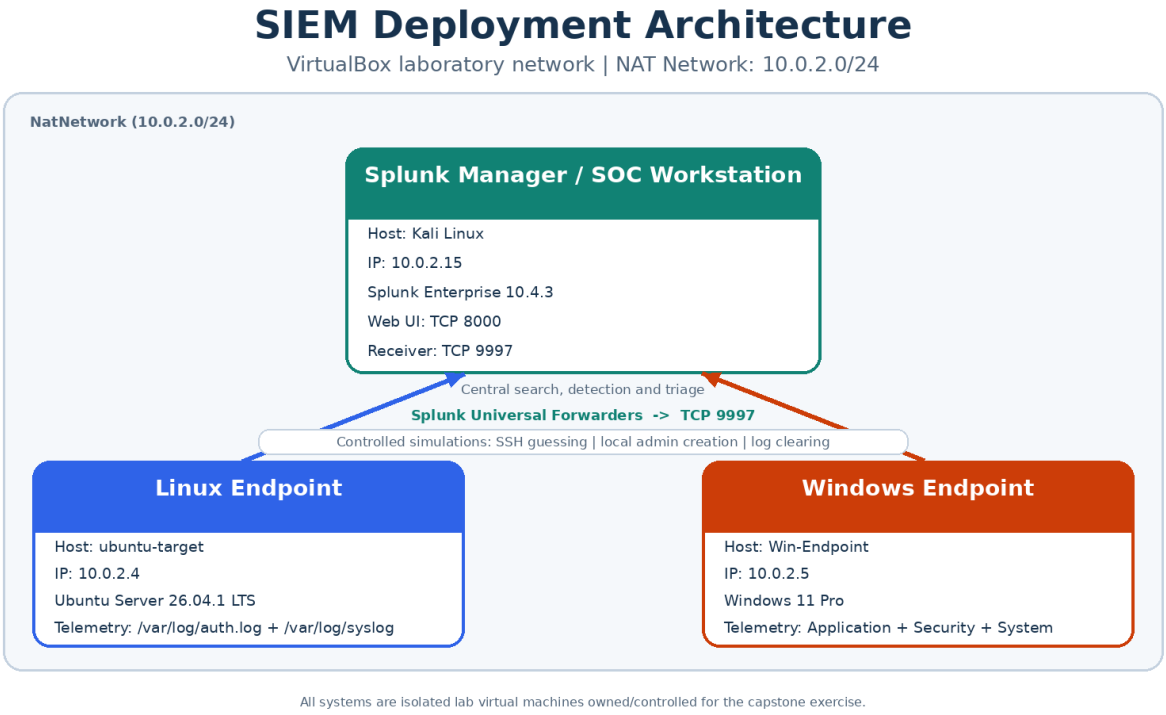



***Figure 1. SIEM architecture topology for the VirtualBox lab.***

| Component | Role | IP Address | Key Services / Telemetry |
|---|---|---|---|
| Kali Linux | Splunk Manager / SOC workstation | 10.0.2.15 | Splunk Enterprise 10.4.3; Web UI TCP 8000; receiver TCP 9997 |
| Ubuntu Server | Linux endpoint | 10.0.2.4 | SSH service; /var/log/auth.log; /var/log/syslog |
| Windows 11 Pro | Windows endpoint | 10.0.2.5 | Application, Security and System Event Logs |

Architecture note: the lab intentionally combined the Splunk Manager and security workstation on Kali Linux to reduce resource requirements. In a production SOC, the SIEM server and analyst workstation would normally be separated and hardened independently.

## 2. Infrastructure Deployment and Log Ingestion

### 2.1 Splunk Manager

Splunk Enterprise 10.4.3 was installed on Kali Linux and configured as the central manager. The Splunk web interface was made available on TCP port 8000 and a receiving port was configured on TCP 9997 for Universal Forwarder traffic.

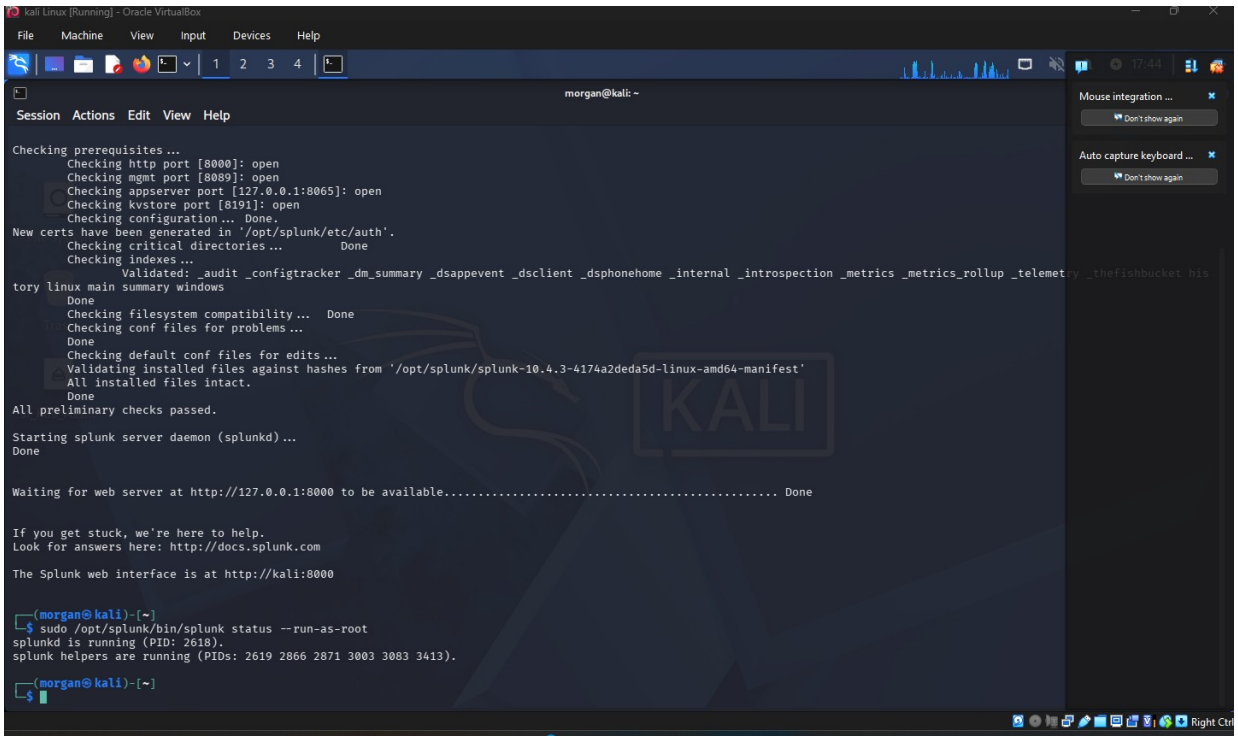



***Figure 2. Splunk Enterprise running successfully on the Kali Linux manager.***

### 2.2 Linux Endpoint

Ubuntu Server 26.04.1 LTS was used as the Linux endpoint with hostname ubuntu-target and IP address 10.0.2.4. The Splunk Universal Forwarder 10.4.3 was installed and connected to 10.0.2.15:9997. The forwarder was configured to monitor /var/log/auth.log and /var/log/syslog. The auth log was assigned a linux_secure sourcetype and syslog used the syslog sourcetype.

```
Forwarding destination: 10.0.2.15:9997
Monitored files:
/var/log/auth.log
/var/log/syslog
```

### 2.3 Windows Endpoint

Windows 11 Pro was deployed as Win-Endpoint with IP address 10.0.2.5. The endpoint was validated on the same NatNetwork and successfully reached the Splunk Manager on TCP port 9997 before installation of the Windows Universal Forwarder.

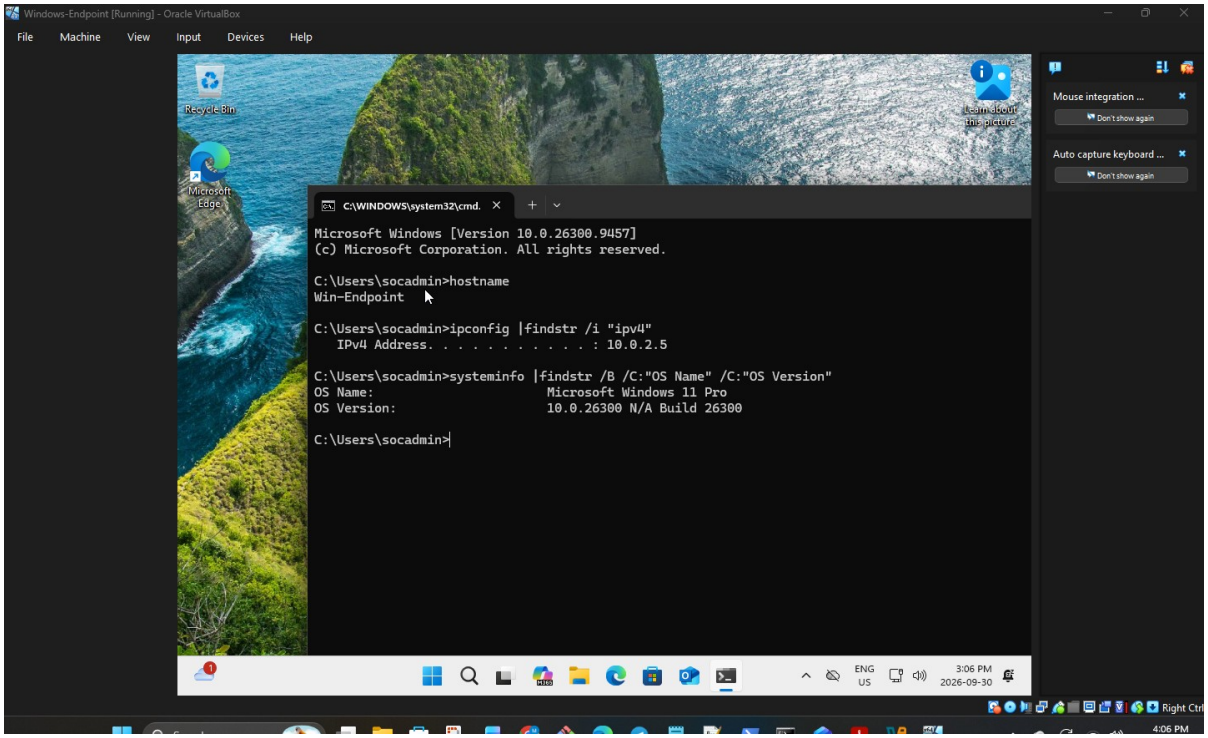



***Figure 3. Windows endpoint identity, IP address and operating-system verification.***

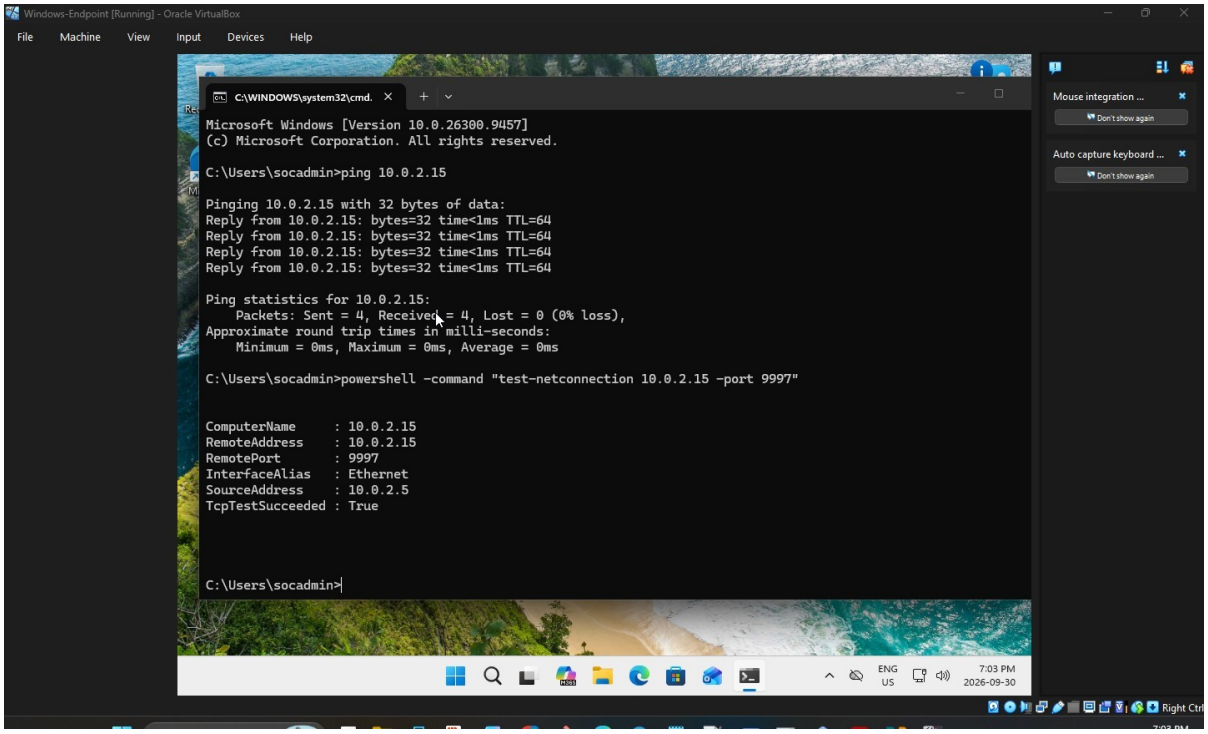



***Figure 4. Windows-to-Splunk network validation: ICMP connectivity and TCP 9997 succeeded.***

Splunk Universal Forwarder 10.4.3 (64-bit Windows MSI) was then installed. The forwarder service was confirmed active, and Application, Security and System Event Logs were selected for forwarding.

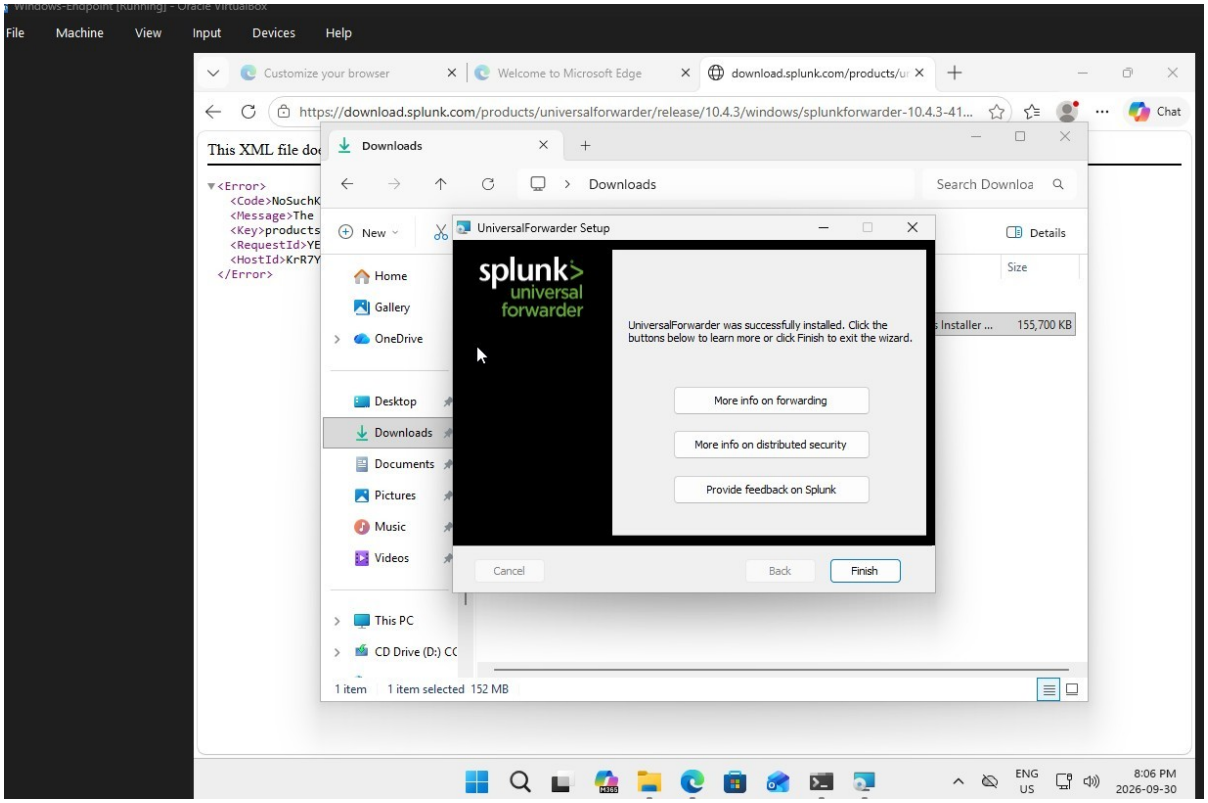



***Figure 5. Splunk Universal Forwarder successfully installed on the Windows endpoint.***

## 3. Proof of Ingestion

### 3.1 Linux Telemetry

Splunk searches confirmed that both required Linux log sources were being parsed from ubuntu-target. At the time of evidence capture, Splunk contained 202 events from /var/log/auth.log and 3,940 events from /var/log/syslog, for 4,142 Linux events in the displayed result set.

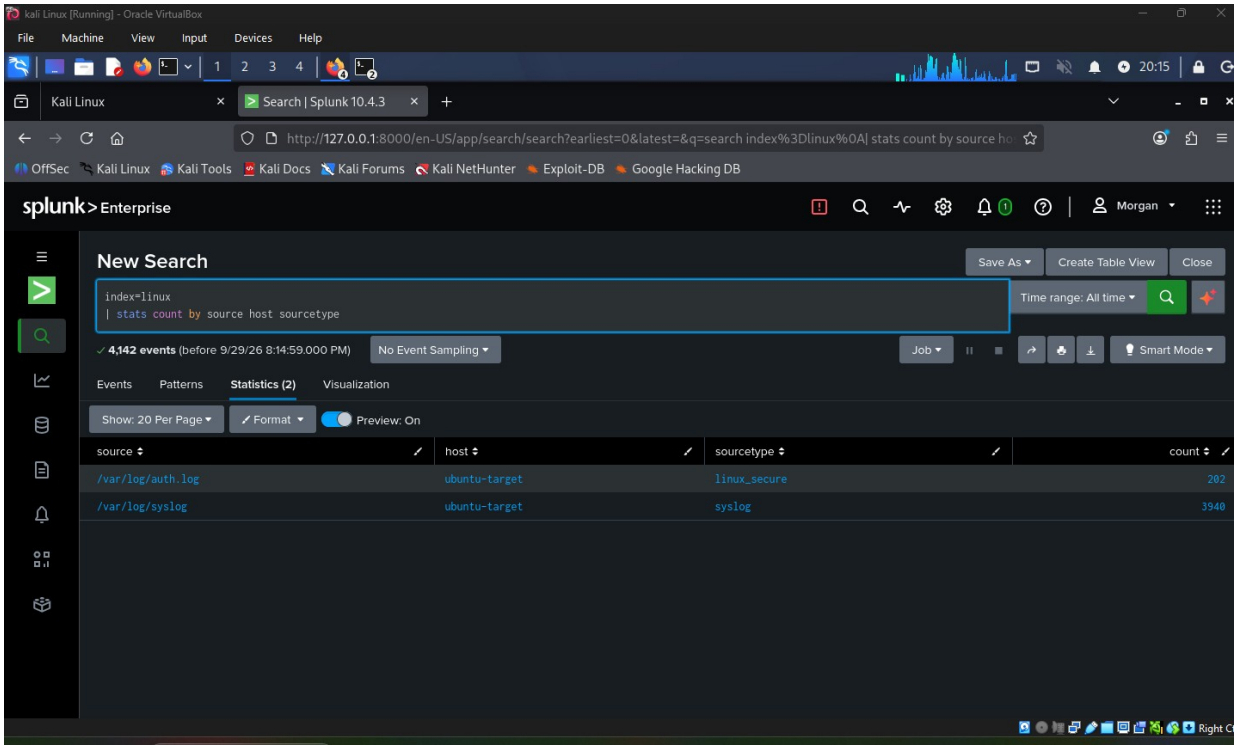



***Figure 6. Proof of Linux ingestion: auth.log and syslog actively parsed in Splunk.***

### 3.2 Windows Telemetry

Splunk also confirmed active ingestion of the three Windows channels required by the project. At the time of evidence capture, 351 Application events, 1,685 Security events and 882 System events were visible, totaling 2,918 events in the displayed search.

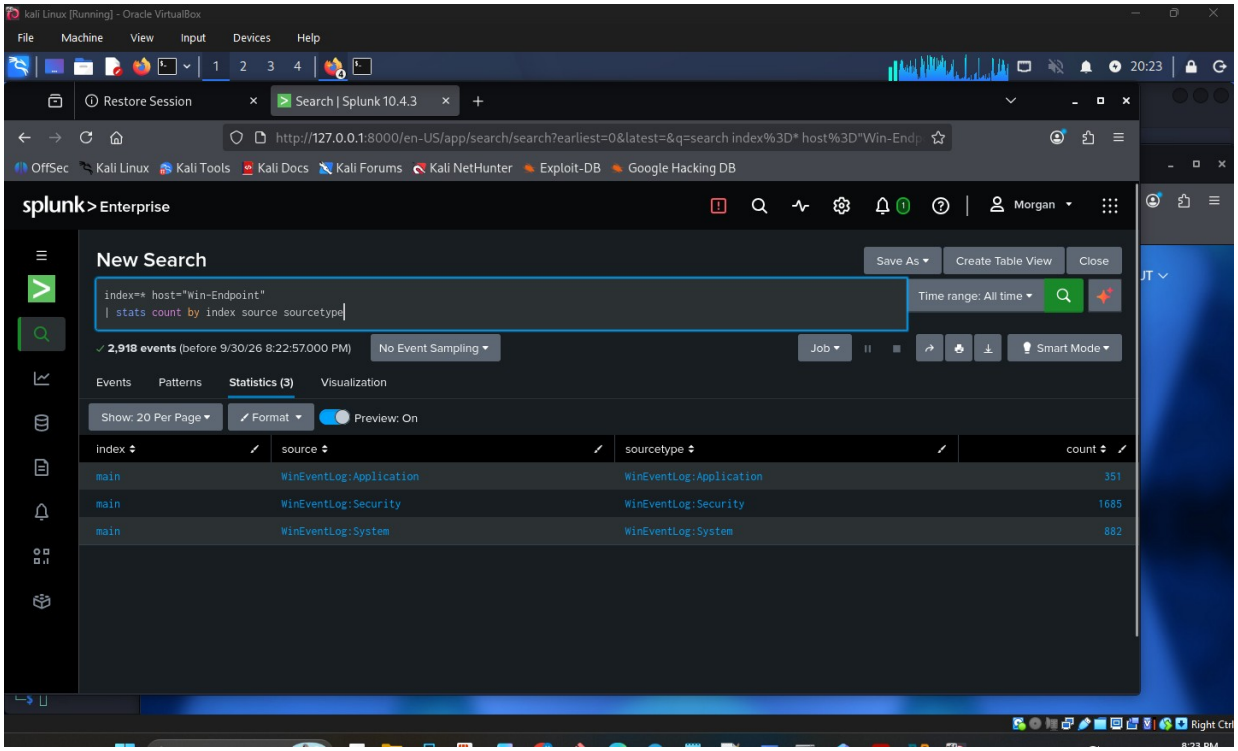



***Figure 7. Proof of Windows ingestion: Application, Security and System event logs in Splunk.***

**Proof-of-ingestion status**
Linux requirement: PASSED. Windows requirement: PASSED. Both endpoints were centrally searchable in the Splunk Manager.

## 4. Controlled Attack Simulation

Three attack vectors were selected from the capstone requirements. Each test was executed against an owned lab endpoint, followed by a Splunk search to verify that the associated telemetry had been ingested.

### 4.1 Attack 1 - Unauthorized Local Administrator Account

A temporary account named siemtest1 was created on Win-Endpoint from an elevated local session, added to the local Administrators group, and hidden from the normal Windows sign-in user list through the SpecialAccounts\UserList registry key. The account was used only as a controlled simulation artifact.

```
net user siemtest1 * /add
net localgroup Administrators siemtest1 /add
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon\
SpecialAccounts\UserList" /v siemtest1 /t REG_DWORD /d 0 /f
```

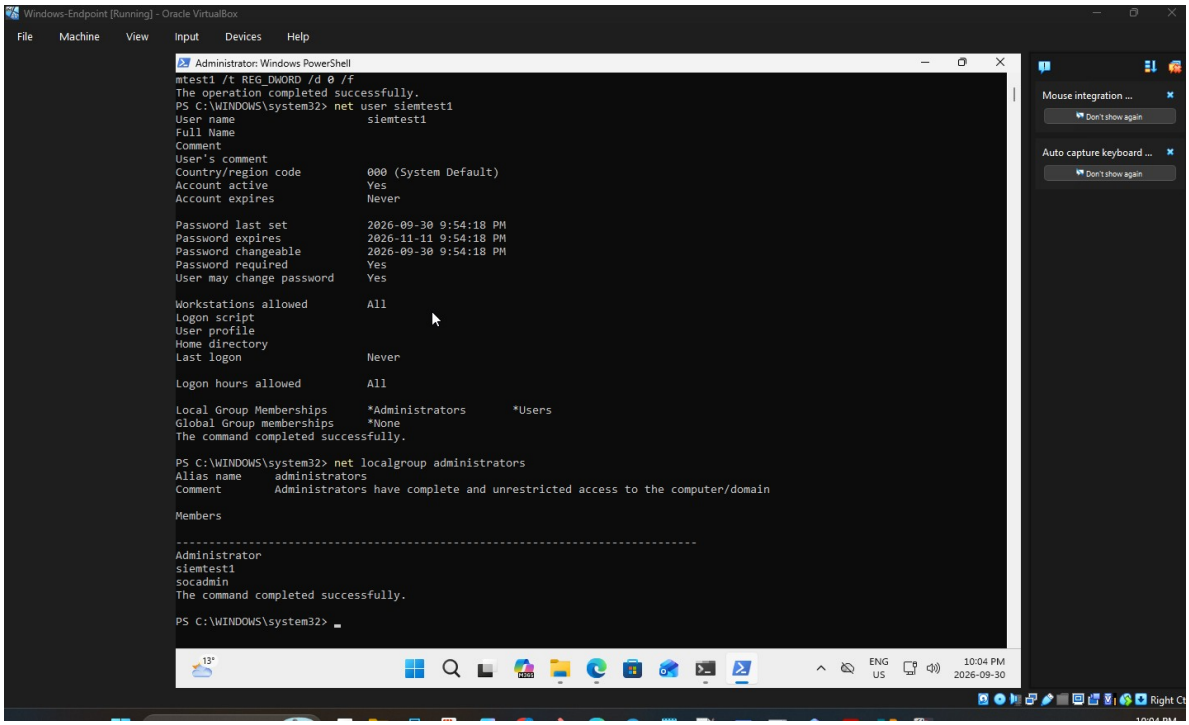



***Figure 8. Attack evidence: siemtest1 exists and is a member of the local Administrators group.***

Splunk returned Windows Security telemetry containing the siemtest1 account. This confirmed that the account-management activity was visible to the SIEM.

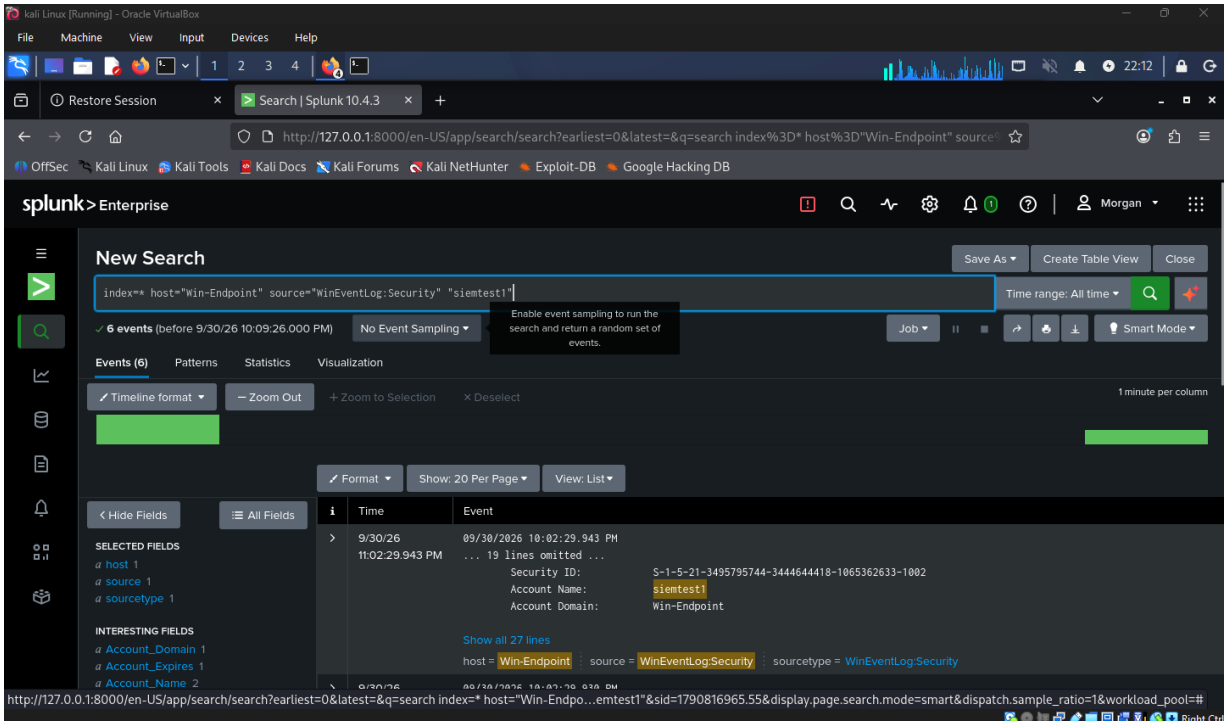



***Figure 9. Splunk detection of Windows Security events associated with siemtest1.***

### 4.2 Attack 2 - Localized SSH Password Guessing

From the Kali security workstation, repeated incorrect passwords were submitted to the Ubuntu SSH service for the user [username]. The SSH server rejected the attempts and recorded them in /var/log/auth.log.

```
ssh [username]@10.0.2.4
# Deliberately enter incorrect passwords during the controlled lab test.
```

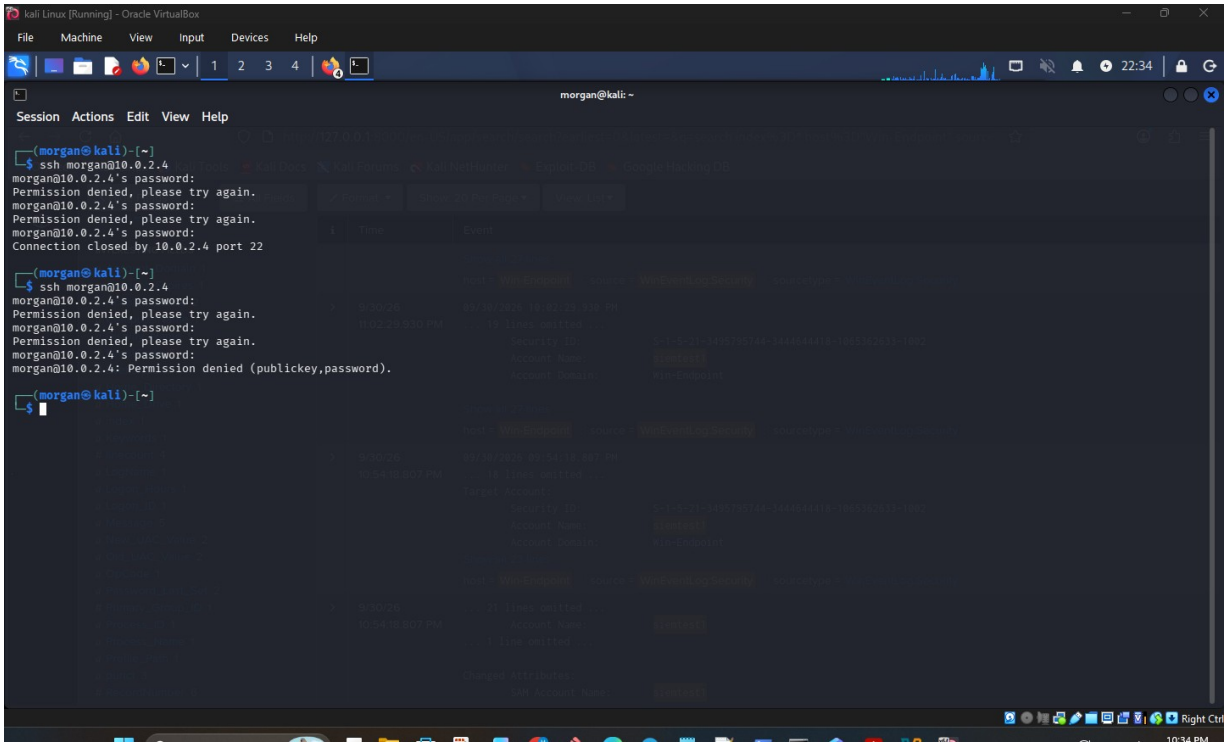



***Figure 10. Attack evidence: repeated failed SSH authentication attempts from Kali to Ubuntu.***

Splunk identified nine failed-password events from source IP 10.0.2.15 against ubuntu-target, demonstrating successful detection of the repeated authentication failures.

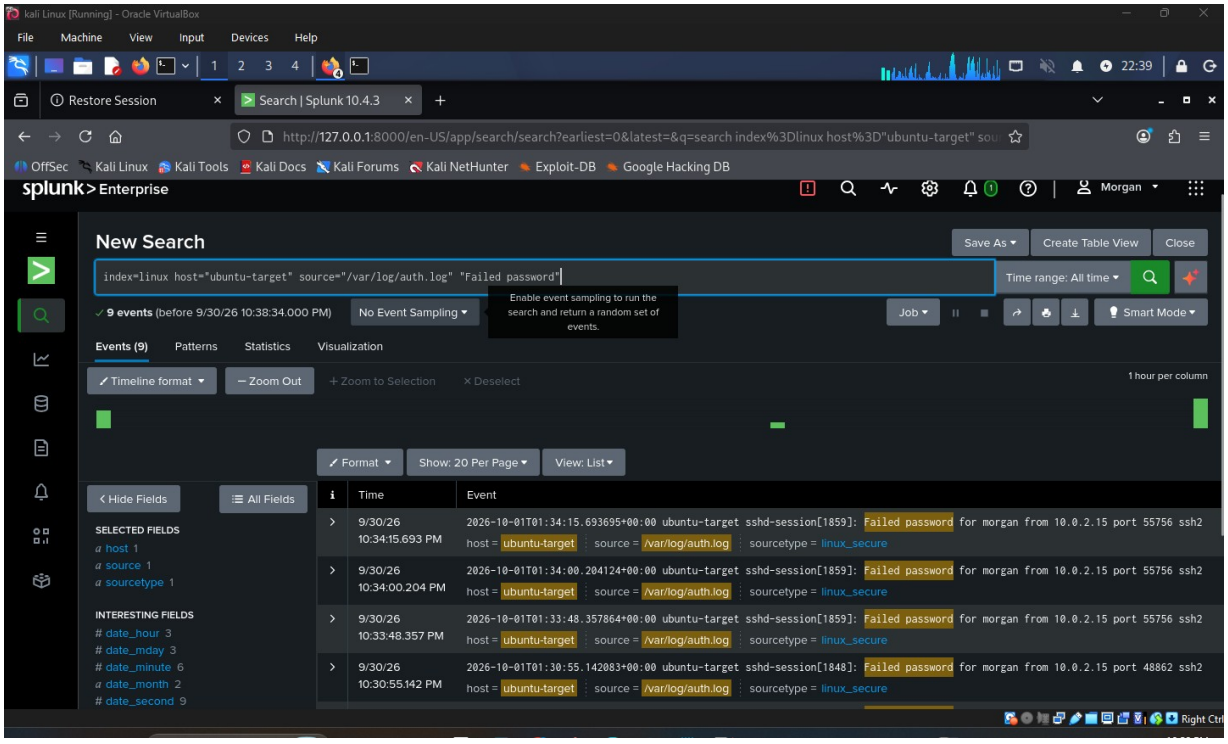



***Figure 11. Splunk detection of repeated SSH failed-password events in /var/log/auth.log.***

### 4.3 Attack 3 - Clearing the Windows Security Event Log

From an elevated Windows command prompt, the Security event log was cleared using the native wevtutil utility. This action is a classic attempt to impair or remove local forensic visibility. The centralized Splunk copy of earlier events remained available.

```
wevtutil cl Security && echo Security log cleared successfully
```

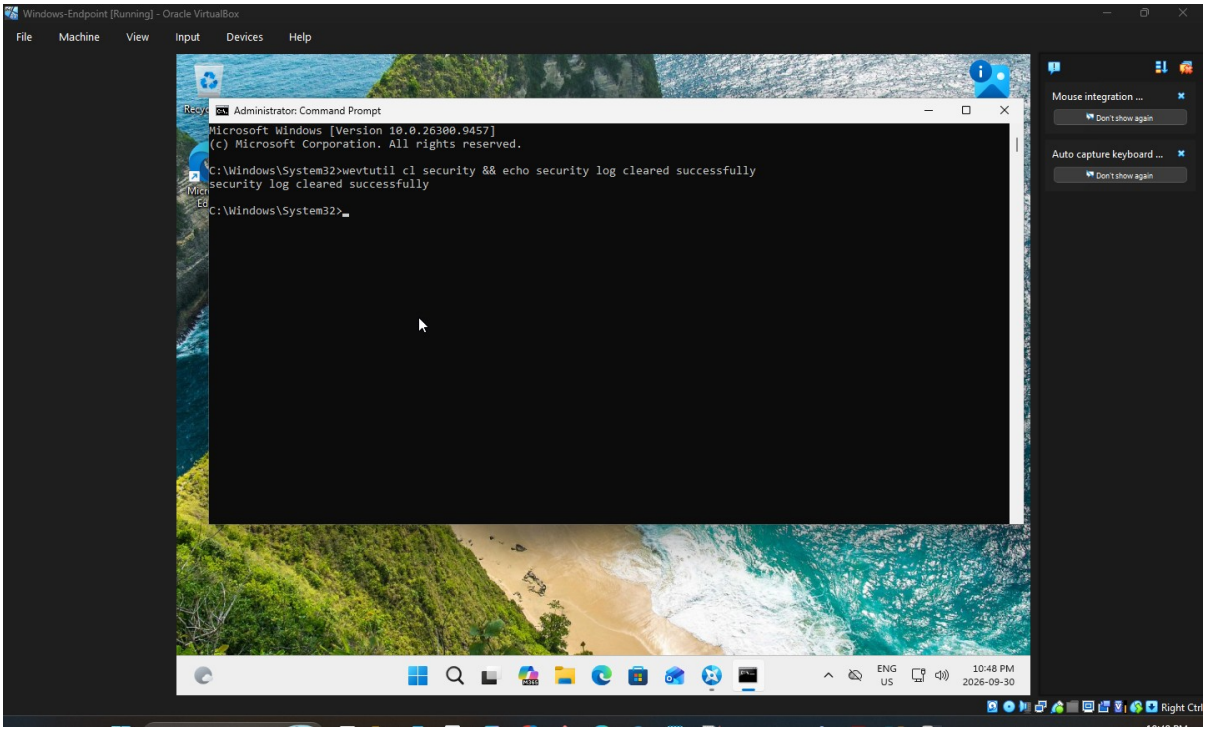



***Figure 12. Attack evidence: the Windows Security log was cleared successfully.***

Windows generated Security Event ID 1102 for the audit-log clearing action, and the event was forwarded to Splunk.

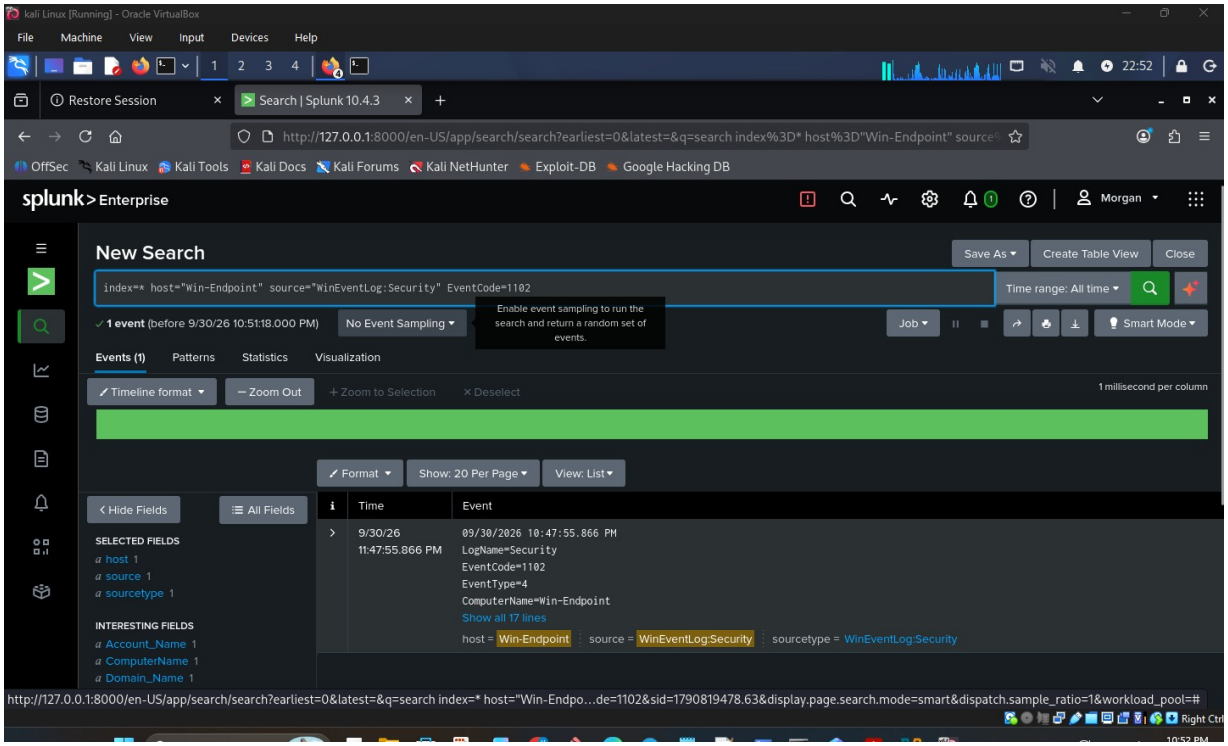



***Figure 13. Splunk detection of Windows Security Event ID 1102 after audit-log clearing.***

### 4.4 Attack Summary

| Scenario | Observed Telemetry | MITRE ATT&CK Mapping | Result |
|---|---|---|---|
| Unauthorized local admin account | Windows Security Event 4720; local Administrators membership | T1136.001 Create Account: Local Account; T1098.007 Additional Local or Domain Groups | Detected |
| SSH password guessing | Repeated 'Failed password' records in /var/log/auth.log | T1110.001 Password Guessing (Credential Access) | Detected |
| Clear Windows Security log | Windows Security Event 1102 | T1685.005 Clear Windows Event Logs (Defense Impairment, current ATT&CK 2026) | Detected |

MITRE note: current ATT&CK (2026) maps Clear Windows Event Logs to T1685.005 under Defense Impairment. Older ATT&CK releases used T1070.001 for this behavior.

## 5. Detection Engineering and Rule Tuning

The attack telemetry was converted into repeatable Splunk searches suitable for SOC triage. The SSH rule was threshold-tuned to reduce noise by requiring at least five failures within a five-minute time bucket from the same source IP and username.

### 5.1 Tuned SSH Brute-Force Detection

```
index=linux host="ubuntu-target" source="/var/log/auth.log" "Failed password"
| rex "Failed password for (?:invalid user )?(?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\
d{1,3}){3})"
| bin _time span=5m
| stats count as failed_attempts by _time src_ip username
| where failed_attempts >= 5
```

The tuned search produced a single detection for source 10.0.2.15, username [username], with five failed attempts in the same five-minute window. This tuning suppresses isolated user mistakes while preserving a clear brute-force signal.

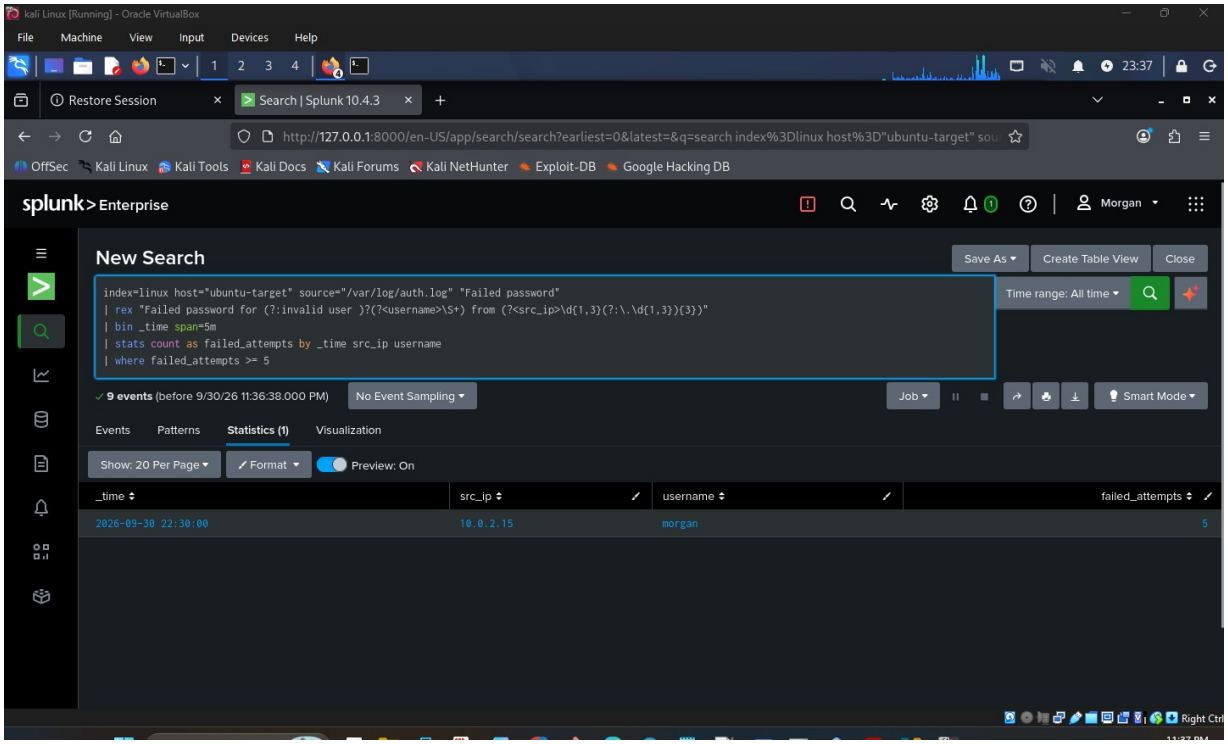



***Figure 14. Tuned SSH brute-force rule detecting five failures in a five-minute window.***

### 5.2 Local Account / Administrator Detection

```
index=* host="Win-Endpoint" source="WinEventLog:Security" "siemtest1"
(EventCode=4720 OR EventCode=4732)
| eval detection=case(EventCode=4720,"Local user account
created",EventCode=4732,"User added to local security group")
| table _time EventCode detection Message
```

The final evidence captured Event ID 4720 for siemtest1 and displayed the full account-creation message. The rule also covers Event ID 4732 so that additions to local security groups can be correlated when present.

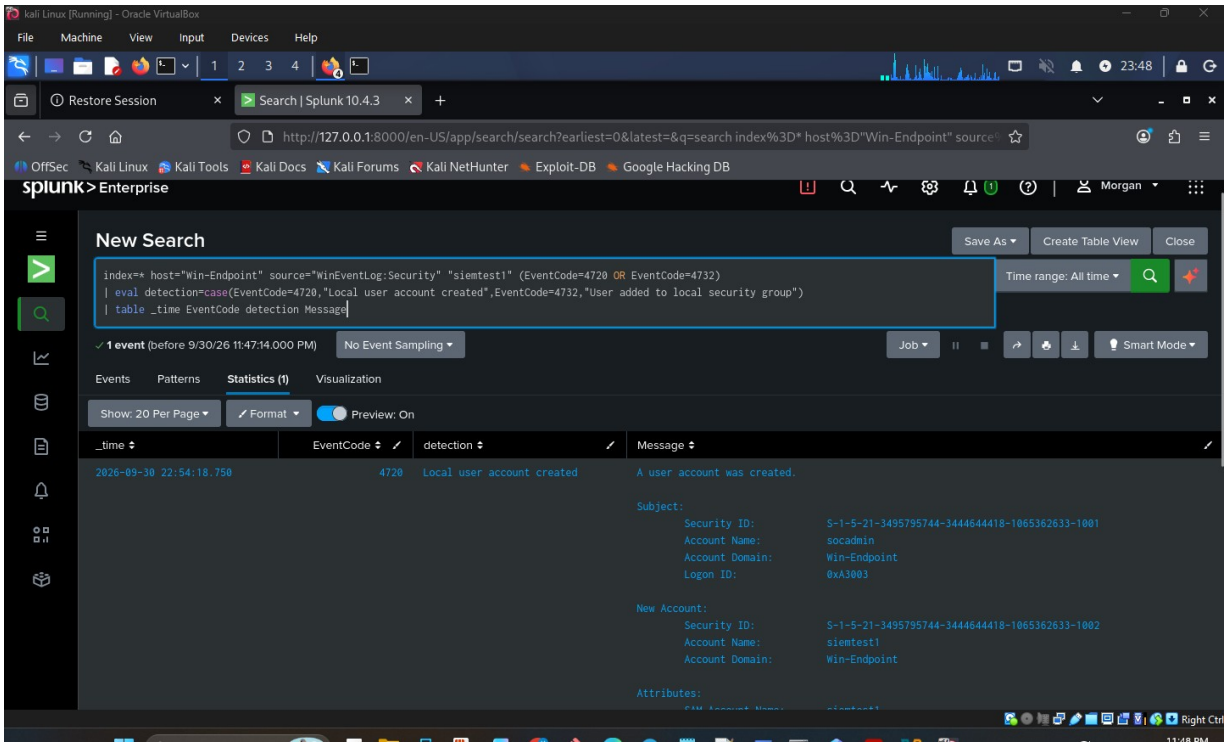



***Figure 15. Detection rule identifying Event ID 4720 for creation of siemtest1.***

### 5.3 Security Log-Clearing Detection

```
index=* host="Win-Endpoint" source="WinEventLog:Security" EventCode=1102
| eval detection="Windows Security audit log cleared"
| table _time host EventCode detection Message
```

Event ID 1102 is high-value security telemetry and should generally be treated as suspicious unless the action is associated with an approved maintenance procedure.

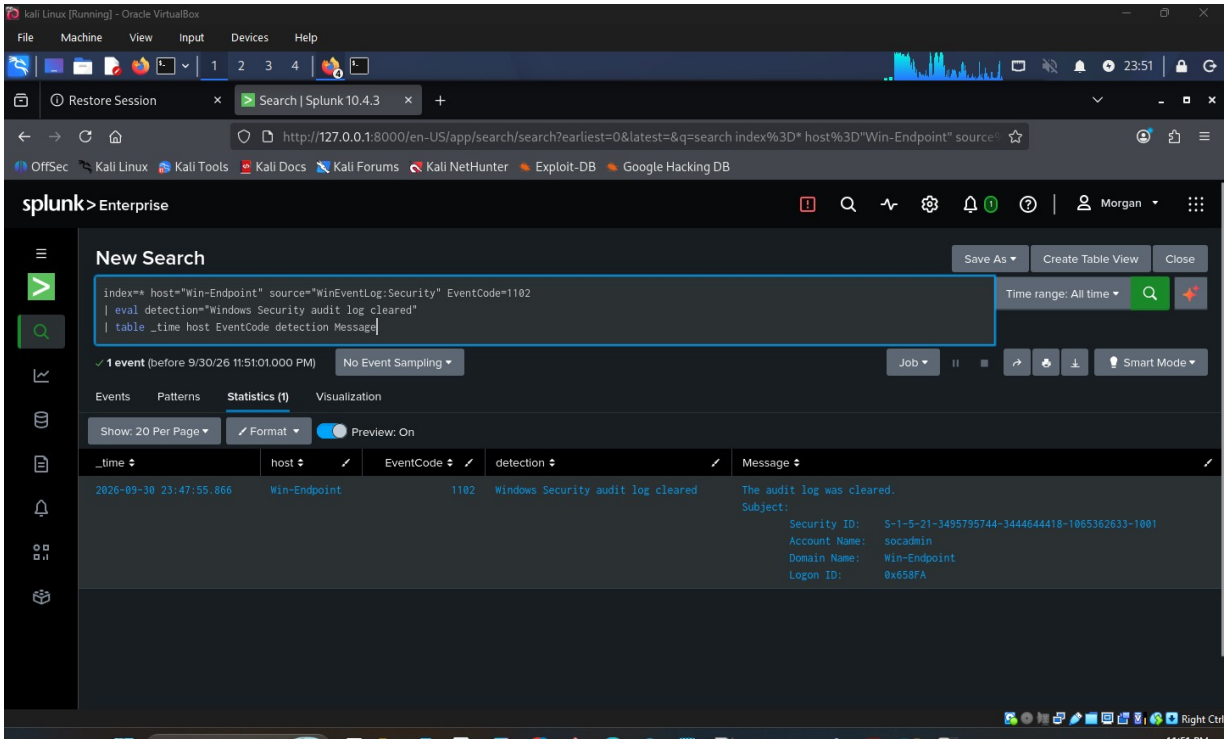



***Figure 16. Detection rule identifying Windows Security Event ID 1102.***

## 5.4 Rule-Tuning Rationale

| Detection | Tuning Decision | Reason |
|---|---|---|
| SSH brute force | 5 or more failures in 5 minutes, grouped by source IP and username | Reduces alerts caused by isolated typing mistakes while identifying repeated guessing. |
| Account creation | Alert on 4720; correlate 4732 for privileged group membership | New local accounts are relatively infrequent and high-value on managed endpoints. |
| Security log clear | Alert on every 1102 | Audit-log clearing is rare and directly reduces forensic visibility; false-positive tolerance should be low. |

## 6. Formal Incident Response Report

### 6.1 Incident Identification

| Field | Value |
|---|---|
| Incident Title | Unauthorized Creation of a Privileged Local Account |
| Affected Asset | Win-Endpoint (10.0.2.5) |
| Test Account | siemtest1 |
| Primary Data Source | WinEventLog:Security |
| Key Event | Event ID 4720 - A user account was created |
| Primary ATT&CK Mapping | T1136.001 - Create Account: Local Account |
| Secondary ATT&CK Mapping | T1098.007 - Additional Local or Domain Groups |
| Primary Tactic | Persistence |
| Secondary Tactic | Persistence / Privilege Escalation |
| Disposition | Confirmed controlled security simulation |

### 6.2 Incident Summary

During the controlled exercise, an elevated local Windows session was used to create a new account named siemtest1. The account was then added to the local Administrators group and hidden from the normal sign-in list. The activity represented a realistic persistence mechanism because a newly created privileged account could provide an attacker with a secondary means of access.

MITRE ATT&CK maps local-account creation to T1136.001 (Create Account: Local Account). ATT&CK also maps addition of an account to a local privileged group to T1098.007 (Additional Local or Domain Groups). The latter carries Persistence and Privilege Escalation tactics. [1][2]

### 6.3 Raw Log Evidence

The following fields were captured in Splunk from the Windows Security log for the simulated account-creation event:

```
TimeCreated: 2026-09-30 22:54:18.750
EventCode: 4720
Message: A user account was created.

Subject:
  Account Name: socadmin
  Account Domain: Win-Endpoint
  Logon ID: 0xA3003

New Account:
  Account Name: siemtest1
  Account Domain: Win-Endpoint
```

Microsoft documents Event ID 4720 as 'A user account was created,' and MITRE's current detection strategy for T1136.001 explicitly identifies WinEventLog:Security EventCode=4720 as relevant user-account creation telemetry. [1][5]

**6.4 Incident Timeline**

| Stage | Time / Order | Observed Activity | Evidence |
|---|---|---|---|
| Assumed initial access | Start of simulation | Adversary access was intentionally assumed: the test began from an already elevated local session. No external exploit was simulated. | Lab scope / controlled exercise |
| Execution | 22:54:18 (Splunk event time) | Local account siemtest1 was created with Windows account-management tooling. | Security Event 4720; Figure 15 |
| Privilege / persistence | Immediately afterward | siemtest1 was added to the local Administrators group. | Local group verification; Figure 8 |
| Concealment | Immediately afterward | Registry entry under SpecialAccounts\ UserList was set to hide siemtest1 from the normal sign-in list. | Successful registry command; Figure 8 |

| Stage | Time / Order | Observed Activity | Evidence |
|---|---|---|---|
| Detection | Within the forwarding interval | Windows Security telemetry was forwarded to Splunk and the siemtest1 account activity became searchable. | Figure 9 and Figure 15 |
| Triage | SOC analysis phase | The event was classified as confirmed suspicious account creation within the simulation and mapped to ATT&CK techniques. | Formal incident analysis |

Timestamp note: the lab used multiple virtual machines with different displayed timezone contexts. The timeline therefore uses the Splunk event timestamp for the exact account-creation event and relative sequencing for subsequent actions.

### 6.5 Impact Assessment

If this activity occurred outside an approved administration workflow, the principal risk would be persistent privileged access. An attacker-controlled local administrator could install software, alter security configuration, access locally available data, create additional accounts, disable controls, or perform further defense-impairment actions.

### 6.6 Incident Containment and Eradication

- Preserve relevant logs and forensic artifacts before cleanup, including the centralized Splunk copy.
- Disable the suspicious account immediately, remove it from privileged groups, and delete it after evidence preservation.
- Remove any SpecialAccounts\UserList registry entry used to hide the account.
- Review the initiating administrator account (socadmin in the lab evidence) and rotate credentials if compromise is suspected.
- Search for related Windows events such as 4720, 4732, 4738, 4726 and, where enabled, process-creation telemetry that identifies net.exe, PowerShell, or other account-management tools.
- Check for additional persistence mechanisms, remote-login activity, scheduled tasks, services, startup items, and changes to security tooling.

### 6.7 Technical Recommendations

- Apply privileged-access management: minimize standing local administrators and require separate administrative accounts for elevated tasks.
- Use Windows LAPS or equivalent managed local-administrator password rotation to prevent password reuse and reduce the value of compromised local credentials.
- Maintain Account Management auditing and centrally forward Security events so local deletion or log clearing does not remove the SOC copy.

- Alert on every unexpected Event ID 4720 and correlate it with Event ID 4732 when the new account is added to Administrators or another privileged group.
- Establish an approval/change window for legitimate account provisioning so SIEM rules can suppress known automation while prioritizing out-of-window changes.
- Where technically applicable, require MFA for remote and privileged access paths and enforce strong password/account-use policies.

### 7. Technical Remediation Recommendations

The following recommendations address weaknesses demonstrated by the three simulations and improve both prevention and detection.

| Area | Recommendation | Priority |
|---|---|---|
| Windows privileged accounts | Restrict local administrator membership; deploy LAPS; use separate admin identities; audit 4720/4732/4738. | High |
| Linux SSH | Prefer SSH keys, disable unnecessary password authentication, restrict source networks, and apply sensible account lockout/rate limiting. | High |
| Central logging | Continue forwarding to a remote SIEM with minimal delay so endpoint log clearing cannot erase the central evidence copy. | High |
| Log-clear monitoring | Treat Event 1102 as a high-priority alert; investigate the subject account and adjacent process/activity telemetry. | High |
| Brute-force detection | Operationalize the >=5 failures/5 minutes rule, then tune thresholds based on normal environment behavior. | Medium-High |
| Time synchronization | Use consistent NTP/timezone configuration across manager and endpoints to simplify incident timelines and correlation. | Medium |
| Change management | Maintain approved windows and service-account allowlists for legitimate account creation and group modification. | Medium |
| Endpoint hardening | Keep operating systems, OpenSSH and Splunk components patched; restrict administrative tools to authorized users. | Medium |

### 7.1 Attack-Specific Recommendations

SSH password guessing:

- Enforce strong unique credentials and prefer key-based SSH authentication.
- Use account lockout or rate-limiting controls carefully to reduce password guessing without creating an easy denial-of-service condition.
- Limit SSH exposure with host firewall rules or management-network allowlists.

Unauthorized privileged account creation:

- Restrict who can create users and modify Administrators-group membership.
- Use SIEM correlation to link 4720 account creation with 4732 privileged-group addition within a short time window.
- Review unexpected hidden-account registry entries as potential persistence artifacts.

Windows event-log clearing:

- Keep Security logs forwarded to an external repository and alert immediately on Event 1102.
- Limit administrative privileges and investigate any use of wevtutil to clear Security, System or Application logs.
- Protect local event-log files and logging services from unnecessary modification.

## 8. Conclusion

The capstone successfully established a functional SIEM environment from scratch. Splunk Enterprise centrally received Linux and Windows telemetry, the required endpoint log sources were demonstrably searchable, and three controlled attack vectors produced detectable security events. Detection engineering converted those raw events into repeatable SOC searches, including a tuned SSH brute-force rule that reduced noise by applying a five-failures-in-five-minutes threshold.

The exercise also demonstrated the defensive value of centralized logging: even when the Windows Security log was cleared locally, Splunk retained the Event ID 1102 evidence. The formal incident analysis of the siemtest1 account creation further showed how endpoint telemetry, ATT&CK mapping, timeline reconstruction, and remediation recommendations can be combined into a practical SOC workflow.

## 9. References

[1] MITRE ATT&CK. T1136.001 - Create Account: Local Account. https://attack.mitre.org/techniques/T1136/001/

[2] MITRE ATT&CK. T1098.007 - Additional Local or Domain Groups. https://attack.mitre.org/techniques/T1098/007/

[3] MITRE ATT&CK. T1110.001 - Brute Force: Password Guessing. https://attack.mitre.org/techniques/T1110/001/

[4] MITRE ATT&CK. T1685.005 - Clear Windows Event Logs. https://attack.mitre.org/techniques/T1685/005/

[5] MITRE ATT&CK. DET0447 / AN1235 - Local Account Creation Across Platforms; WinEventLog:Security EventCode=4720. https://attack.mitre.org/detectionstrategies/DET0447/

[6] Microsoft Learn. Event 1102(S): The audit log was cleared. https://learn.microsoft.com/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/event-1102

[7] Microsoft Learn. Advanced Audit Policy Configuration settings (Account Management event IDs, including 4720). https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/plan/security-best-practices/advanced-audit-policy-configuration

**Evidence basis**
All screenshots embedded in this report were captured during our SIEM capstone lab. Event counts are point-in-time values from the evidence screenshots and may increase as the forwarders continue to send telemetry.